



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -1 (Paper-2)

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C Programming

2. Iterative Control Instruction

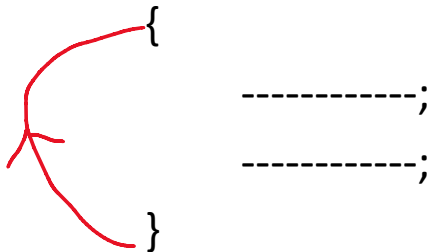
An **iterative control loop** is a loop that repeats a set of statements again and again until a condition becomes false.

Example

- a. While
- b. do while
- c. for

a. While

Syntax → while (condition)



Q. WAP to print “Hello Bihar” ten times on the screen.

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
int main()
```

```
{
```

```
int x=1;
```

```
while (x<=10)
```

```
{
```

```
printf("\n Hello Bihar");
```

```
    x++; //x=x+1;
```

```
}  
    getch();  
}
```

OUTPUT

```
Hello Bihar  
Hello Bihar  
Hello Bihar  
Hello Bihar  
Hello Bihar  
Hello Bihar  
Hello Bihar  
Hello Bihar  
Hello Bihar  
Hello Bihar
```

b. do-while

Syntax → do

```
{  
    -----;  
    -----;  
} while (condition);
```

Q. WAP to print 1 to 10 natural number using do-while loop.

```
#include<stdio.h>  
#include<conio.h>  
int main()  
{  
    int x=1;  
    do  
    {  
        printf("\n %d ",x);
```

```
x++;  
} while (x<=10);  
getch();  
}
```

OUTPUT

1 2 3 4 5 6 7 8 9 10

c. for

Syntax : → for(-----;-----;-----)
{
 -----;
 -----;
}

Q. WAP to print 1 to 10 natural number using for loop.

```
#include<stdio.h>  
#include<conio.h>  
int main()  
{  
    int x;  
    for (x=1;x<=10;x++)  
    {  
        printf("%d ",x);  
    }  
    getch();  
}
```

OUTPUT

1 2 3 4 5 6 7 8 9 10

All question written in all three loop condition(for, do-while, while).

1. Print numbers from 1 to N
2. Print numbers from N to 1 (reverse order)

3. Print even numbers from 1 to N
4. Print odd numbers from N to 1
5. Find sum of first N natural numbers
6. Print table of 5 in horizontal / portrait
7. Print table of 5 in Vertical / Landscape
8. Find factorial of a number
9. Print multiplication table of a number
10. Count digits in a number
11. Reverse a given number
12. Check whether a number is prime or not

❖ Jump Control Instruction

A **jump control instruction** is a control statement that transfers program control from one part of the program to another part.

Example –

1. break
2. continue
3. goto
4. return

1. Break

- Break is a keyword
- Break can be used either in loop's body or switch's body.
- In loop, break is used to terminate execution of loop.

Q. Write a program in c when user press even number then won the match else loss/game over. A total three chances are given.

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
int main()
```

```
{
```

```
    Int x,i;
```

```
    For(i=1;i<=3;i++)
```

```
    {
```

```

        Printf("Enter an even number");
        Scanf("%d", &x);
        If (x%2==0)
            Break;
    }
    If (i==4)
        Printf("game over");
    else
        printf("Win game");
}

```

CW/HW

Q. WAP in C language create only one MCQ type question and check this.total chance 3 times.

2. Continue

continue is a jump control instruction in C language that skips the current iteration of a loop and moves to the next iteration.

Example –

```

#include<stdio.h>
Int main()
{
    Int i;
    for(i=1; i<=5; i++)
    {
        if(i==3)
            continue;

        printf("%d", i);
    }
}

```

```
}
```

Output

```
1 2 4 5
```

CW/HW

Q1. WAP to print Only Even Number to 1 to 50.

Q2. WAP to print Only Odd number to 50 to 1.

Q3. WAP to check whether the user is eligible for voting or not and 50-70 age groups do breakfast facility and above 70 age group provide vehicle facility.

3. Goto

goto is a jump control instruction in C language that transfers program control to another part of the program using a label.

Syntax

```
goto bottom;
```

bottom:

```
statement;
```

Nested loop and Star pattern Problems

Syntax 1.

```
While()
```

```
{
```

```
    For (---;---;---)
```

```
    {
```

```
        -----;
```

```
        -----;
```

```
    }
```

```
    -----;
```

```
    -----;
```

```
}
```

Syntax2.

```
    for (----;----;----)
```

```
    {
```

```
        for(-----;-----;-----)
```

```
        {
```

```
            -----;
```

```
            -----;
```

```
        }
```

```
        -----;
```

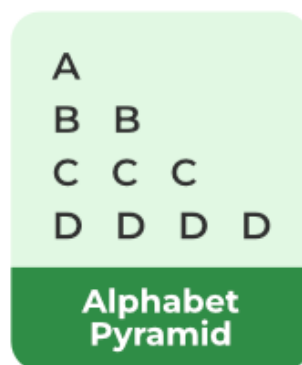
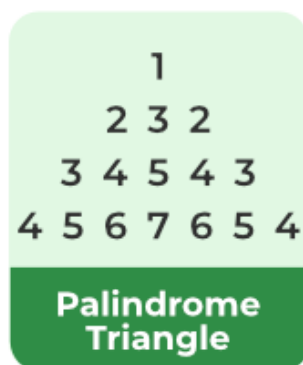
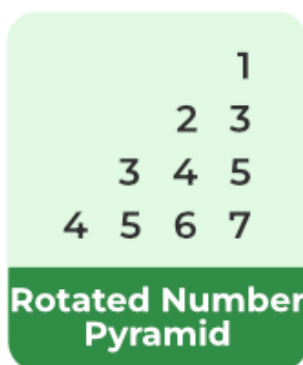
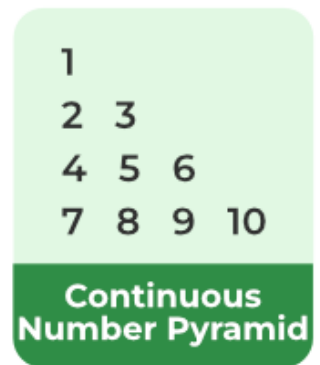
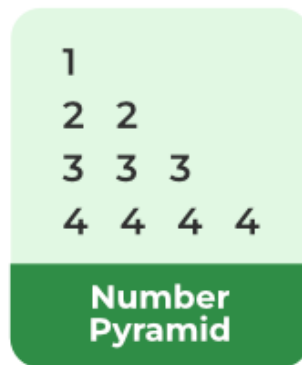
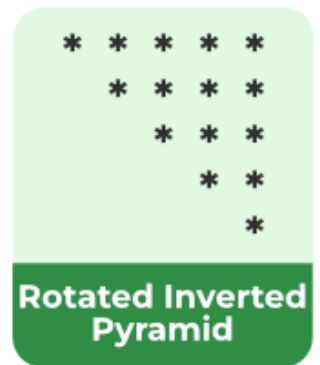
```
        -----;
```

```
}
```

Q22. WAP to generate table chart(1- 10).

1. Print 1 to 10 table in horizontal / portrait
2. Print 1 to 10 table in Vertical / Landscape

Star Pattern / pyramid Pattern



1.Switch case control instruction

Q24. WAP to take input digit by user 0 to 5 (0-5) other input is invalid.